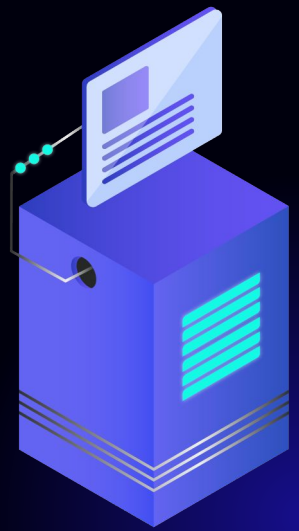
Cloud Workload Execution Models



Gianvito Alcamo, Valerio Triolo, Gabriele Giudici
University of Pisa
Performance Evaluation of Computer Systems and Network Course
A.Y. 2025/2026



UNIVERSITÀ DI PISA



INTRODUCTION

- A Cloud Computing System with finite capacity **C** serving **N** Users
- Each User generates a new Workload every **T** seconds
- Every Workload requires a Computing Capacity **R** for **D** seconds
- If Capacity is not enough → Workload enters a **FCFS Queue**

Two Models

Container Based

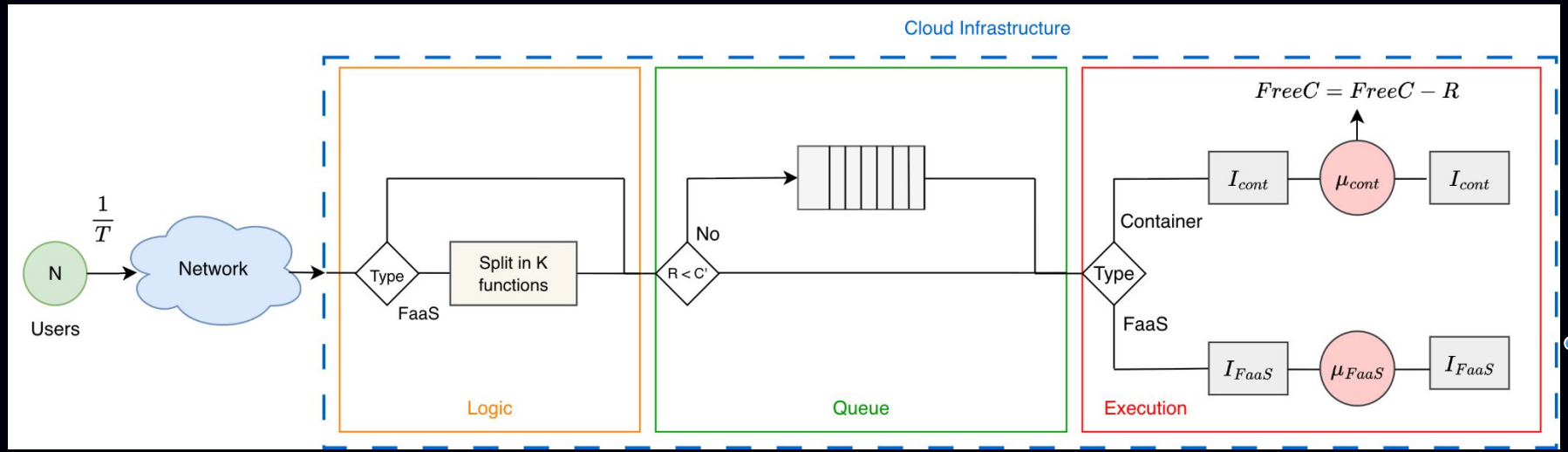
- Monolithic
- Only set up once
- Higher initialization and destruction cost → I_{Cont}

Function as a Service

- Workload gets split into **K** Sequential Functions
- Lower initialization and destruction cost per function → $I_{FaaS} < I_{Cont}$
- Cumulative Overhead

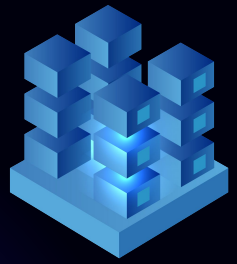
OBJECTIVES

- Evaluating the total execution time of workloads for various values of N and K.
- Comparing the performance between container-based model and Function as a Service model.



Scheme of the System

SYSTEM MODELING

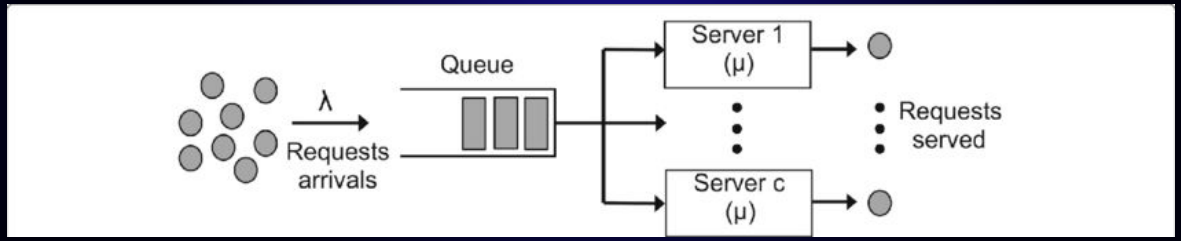


Assumptions

- Capacity **C** is fixed and shared among all the Users
- No Workload loss due to communication channel
- Overheads l_{Cont} and l_{FaaS} do not consume any additional capacity to **R**
- The type of Workload generated by a User is determined using a uniform random distribution

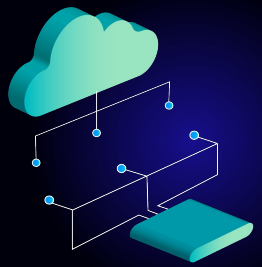
Theoretical Model

- We chose a M/M/c model for our system
- Real model is M/G/c due to overheads
- Real-life capacity **C** is shared among servers



Representation of a M/M/c system

IMPLEMENTATION



VERIFICATION 01

Degeneracy Test

Evaluating the Simulator's robustness under limit conditions

- Idle State and Extreme Saturation ($\lambda \gg \mu$)
- No Capacity and Unlimited Capacity
- Infinite Arrival Rate
- R > C
- Only Container Mode
- Only FaaS Mode
- Results are consistent

Validation through Theoretical model

Checking the correctness of the Simulator's queueing logic

- Used analytical equations for M/M/c systems
- Set overheads to zero ($l_{Cont} = 0$; $l_{FaaS} = 0$)
- Compared Mean Response Time
- R > C
- Theoretical value falls into the confidence interval derived from simulations

VERIFICATION 02

Consistency Test

Evaluating the Simulator's coherence when scaling parameters

- Config. A ⟶ E[T] = 5s ; E[D] = 2.5s
- Config. B ⟶ E[T] = 10s ; E[D] = 5s
- Disabled overheads for pure theoretical evaluation (I_{Cont} = 0 ; I_{FaaS} = 0)
- Utilization and Avg. Queue Length shouldn't change
- Waiting Time and Response Time should double up

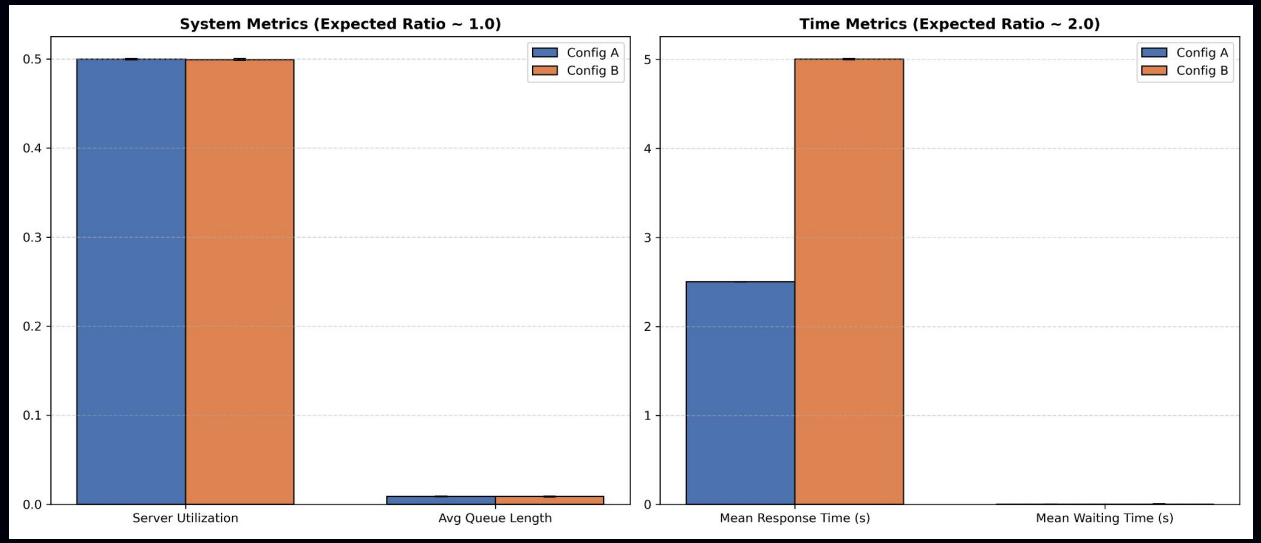
Continuity Test

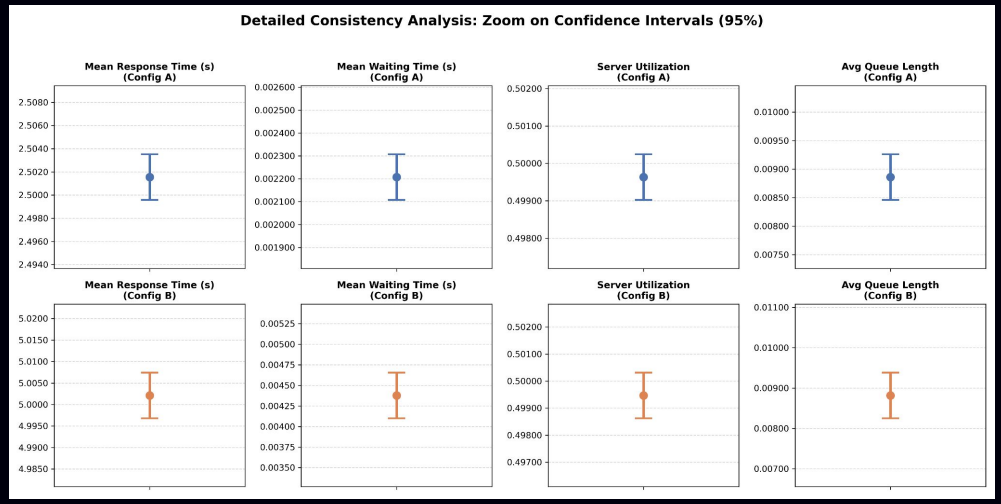
Checking that small input changes result in small predictable output variations

- Medium-High System Load (72%)
- Config. A ⟶ N = 64 Users
- Config. B ⟶ N = 65 Users
- Adopted **Paired Observations** analysis method
- Results proved Continuity and Monotonicity

VERIFICATION

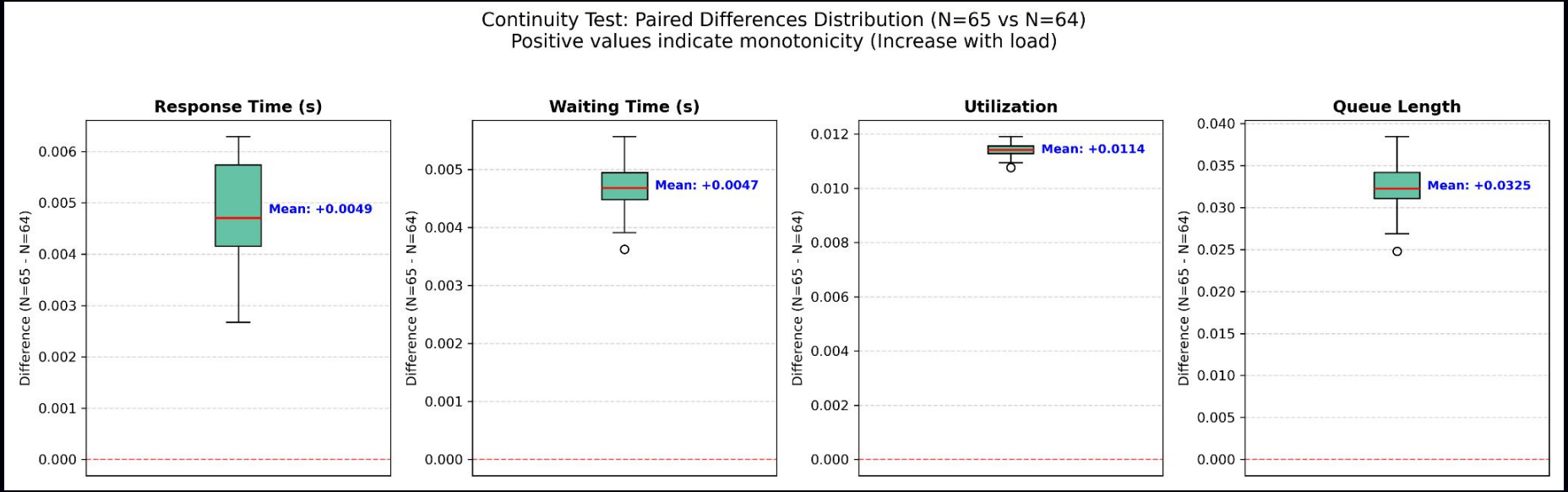
Consistency Test





VERIFICATION

Continuity Test



CALIBRATION

- The Dimensioning of our System was performed by analyzing various scientific papers as references to real-world systems
- Parameters were scaled for practical simulation necessities (scaling factor $\alpha = 1000$)

Parameter	Reference Scenario	Scaled Simulation
Total Capacity (C)	40,000 GHz	40 GHz
Total vCPUs (k)	20,000	20
System Users (N)	[5k ; 100k]	20 ; 90

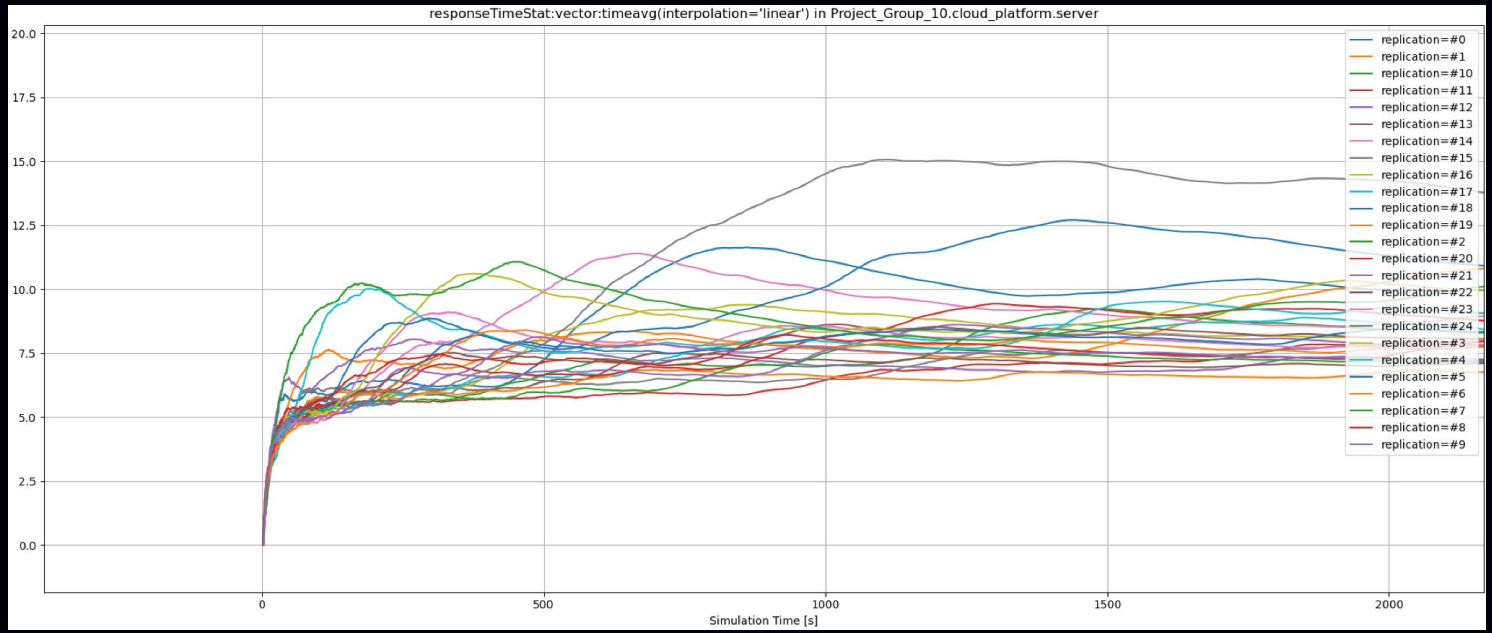
Parameter	Value
l_{Cont}	0.800 s
l_{FaaS}	0.125 s

Overheads

CALIBRATION

Warm-up Time

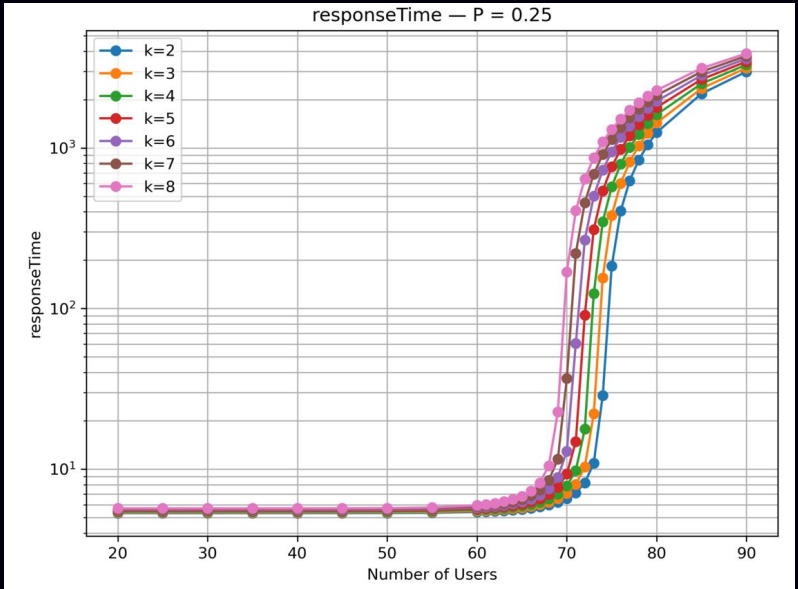
- Calibrated using **Response Time**
- Set to 500 s

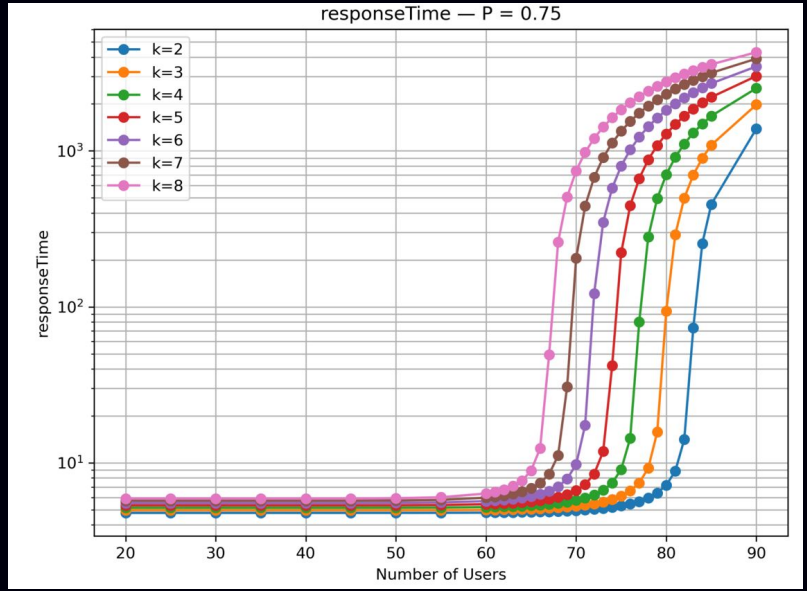


SIMULATION 01

Response Time

- Used different balance between Container-mode and FaaS-mode workloads in the system (p = 0.25 ; p = 0.75)
- Showed the System's behaviour as the number of users increases
- Displayed the impact of different FaaS granularity (**K)**





SIMULATION 02

System's Utilization

- Used different balance between Container-mode and FaaS-mode workloads in the system (p = 0.25 ; p = 0.75)
- Utilization increases linearly, but FaaS workload proportion changes steepness
- System approaches saturation faster when granularity is higher (K)

